

Mountain Lion Detection System

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System Overview

The Mountain Lion Detection System (MLDS) will be used by the San Diego County Parks and Recreation Department to detect mountain lions on land under their jurisdiction. The system will be based on another animal detection system created by the company “Animals-R-Here.” The system should be able to be reconfigured to fit the specific needs of the entity using it, in this case, possibly other parks in the State of California.

Purpose and Scope

The purpose of the MLDS is to detect mountain lions on government land and to generate reports that government employees can use to make informed decisions regarding how to handle the animal. The system provides location data to support ranger decision-making. They can also use this information to analyze data on the local population to determine population size, territory, feeding habits, etc. The system will be able to detect other animals in the area as well.

System Context

The San Diego County Parks and Recreation Department will place audio sensors in locations in their parks. These sensors should work in all outdoor conditions. The sensors will send data to the controlling computer at a park ranger station. The controlling computer will then trigger an alarm that will continue until the ranger turns it off. The ranger on duty will be able to classify the alarm as definite, suspected, or false. The controlling computer will generate a report with relevant data that will be stored in the system in full for 30 days. It will generate a summary for all reports longer than 30 days but shorter than a year.